

# **L1a:** Course Orientation, Markets, and Financial Decisions

CHEME 5660 Cornell University

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# Disclaimer and Risks

**Training only.** This course is for education and information. The teaching team does not offer or endorse any security, derivative, investment, trade, or strategy.

**Risk.** Investing and trading can lose money and may not suit people with limited resources, experience, or tolerance for risk. Past performance does not guarantee future results.

**Responsibility.** You are responsible for decisions based on your finances, goals, tolerance for risk, and need for access to cash.

# Objectives for Today

Today we answer three questions: How does the course work? Why do financial markets matter even if you never become a trader? What makes one investment preferable to another?

## Today's concepts

- **Course roadmap:** four units move from reading payments to making and testing financial decisions.
- **Why markets matter:** most adults and Cornell itself are investors, while companies use public markets to raise money.
- **Our first decision problem:** two investments pay different amounts at different times. How can we compare them?

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**Note:** this lecture has no notebook and no notes packet. These slides are the record.

# How This Course Runs

Across twenty-eight lectures, we move from understanding financial assets to valuing them, combining them in portfolios, and making decisions. Six problem sets use real data.

- **Materials:** download each week from the repository releases page, or clone the repository. After release, files are frozen; corrections are announced.
- **Tools:** Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. Set these up once, at the start of the semester.
- **Questions:** ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

The course ends with an automated trading project using live data from [Alpaca Markets](#).

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**Policies:** [grading](#), [attendance](#), [academic integrity](#), and [accommodations](#) are posted in full

# Ask a Question, Build a Model, Test It

The same cycle appears in every unit.

- **Start with a question.** New concepts appear when they help us solve a financial problem.
- **Use code to explore.** Most code is provided; you will change assumptions, inspect the results, and explain what the model does.
- **Test against evidence.** A mathematically correct model can still fail when its assumptions do not match reality.

The goal is not just to get an answer. It is to know when that answer deserves trust.

# Markets Connect You, Cornell, and Companies

## YOUR SIDE

61 %

of U.S. adults had a retirement account in 2025.

Keeping the default mix is still a choice.

That describes today's adults — not your future.

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## CORNELL'S SIDE

\$10.9B

FY2025 endowment

**\$498M** went into FY2025 operations, mainly from endowment payout.

**\$3.16B** of the endowment was designated for financial aid.

## COMPANY SIDE

### Why go public?

A private company lists shares on a public stock market.

**Raise capital:** sell new shares; the company receives the money.

**Create liquidity:** shareholders can trade existing shares.

**Sources:** [Federal Reserve SHED, 2025](#); [Cornell FY2025 Financial Report](#); [Cornell FY2025 Statement of Activities](#); [U.S. SEC](#)

# U.S. Markets Move About \$1.85 Trillion a Day

To compare annual and daily figures, the daily column divides by 252 trading days. Industry rows show revenue; market rows show cash paid when securities trade.

|                            | Annual total | Per trading day |
|----------------------------|--------------|-----------------|
| <b>Global biopharma</b>    | USD 666B     | USD 2.64B       |
| <b>Global batteries</b>    | USD 181B     | USD 0.72B       |
| <b>Together</b>            | USD 847B     | USD 3.36B       |
| US debt                    | USD 304.7T   | USD 1,209B      |
| US stocks                  | USD 153.2T   | USD 608B        |
| Prices paid for options    | USD 9.3T     | USD 37B         |
| <b>US markets together</b> | USD 467.2T   | USD 1,854B      |

Whether measured per year or per trading day, the market total is about **550 times** the two-industry total. At that scale, small pricing or risk errors can be costly.

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**Sources:** market sizes 2025 (Fortune Business Insights); turnover 2024–2026 (SIFMA, Cboe)

# An Option's Price Is Not the Value It Controls

The options row on the previous slide counts **USD 37B** in premiums: the cash paid to buy options. It does not count the much larger value of the assets tied to those contracts.

Most equity options cover **100 shares**. Across 60.4 million contracts per day, options reference about **USD 4 trillion** of underlying value:

$$\frac{\text{USD 37B}}{\text{price paid (premium)}} \bigg/ \frac{\text{USD 4,000B}}{\text{value controlled}} = \frac{0.92\%}{\text{about 92 cents per \$100}}$$

The **USD 4 trillion** is not cash turnover. It is a reference value: many contracts are cash-settled, expire unused, or close before exercise. That is why the table uses the **USD 37 billion** actually paid.

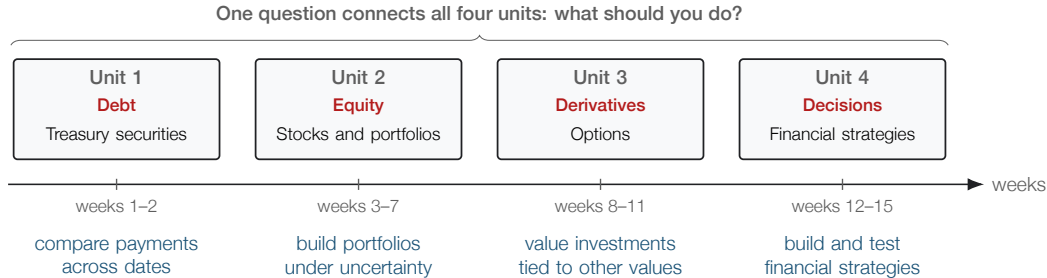
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**Careful:** about USD 3.0T is one cash-settled index; no shares change hands.



# Four Units Build One Decision Toolkit

The first three units teach us to value debt, equity, and derivatives. The fourth uses those tools to build and test financial strategies.



This week starts with Treasury securities.

# Every Unit Uses the Same Decision Loop

Every unit asks three questions: What do we know now? What can we choose? What happens next? Decision models call these the **state**, **action**, and **reward**.



The course gives us tools to describe each part of the loop and test the links between them.

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**Careful:** every arrow represents an assumption we need to test.

# Each Test Exposes a New Way an Idea Can Fail

A model earns trust in stages. Each stage adds evidence—and a new way to fail.

|                                                                    | assumptions | past data | current data | money at risk |
|--------------------------------------------------------------------|-------------|-----------|--------------|---------------|
| <b>Real money</b><br>tests: “I would actually do this”             |             |           |              |               |
| <b>Forward test</b><br>tests: “the market cooperates in real time” |             |           |              |               |
| <b>Backtest</b><br>tests: “the math works on past data”            |             |           |              |               |
| <b>Theory</b><br>tests: “my reasoning is coherent”                 |             |           |              |               |

A **backtest** uses past data. A **forward test** uses current market data, with no money at stake.

Skipping a stage can postpone a problem until real money is at stake.

# Five Promises for \$10,000

You can buy each promise for \$10,000 today. The payments below use market rates from yesterday. Ignore the names for now; compare only what gets paid and when.

- A \$10,028 back, four weeks from today
- B \$236 every six months for 10 years, then \$10,000
- C \$264 every six months for **30** years, then \$10,000
- D \$271 every six months for 10 years, then \$10,000
- E \$121 every six months for 10 years, plus an inflation adjustment on every payment and on the \$10,000

**Which promise would you choose? Vote first; then explain your rule.**

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**As of:** published rates for 2026-08-18, each scaled to a \$10,000 outlay

# What the Five Promises Are Called

|   | What it is called                         | Annual rate       |
|---|-------------------------------------------|-------------------|
| A | 4-week Treasury bill                      | 3.63%             |
| B | 10-year Treasury note                     | 4.71%             |
| C | 30-year Treasury bond                     | 5.28%             |
| D | investment-grade corporate bonds          | 5.42%             |
| E | 10-year inflation-indexed Treasury (TIPS) | 2.41% <b>real</b> |

You compared all five **without knowing any of these names**. The name identifies the kind of promise; the schedule tells you what gets paid and when. Thursday starts from that distinction.

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**Careful:** D is an index of many investment-grade corporate bonds, not one bond. E's rate is stated after inflation, unlike the others.

# The Payment List Leaves Three Questions Open

Choosing the largest listed payment is tempting, but each row hides a different issue.

- **C: What if you need the money before year 30?** Selling early means accepting the market price then, and that price changes with interest rates. Week 2.
- **D: Why does it pay more?** Its rate is 5.42%, versus 4.71% for the ten-year Treasury. That gap reflects the chance of missed payments and the difficulty of selling quickly. Units 2 and 3.
- **E: What does inflation protection change?** Its payments rise with inflation, so its 2.41% rate is not directly comparable with the other rates. Later in the course.

A larger listed payment is not automatically a better investment.

# A Bank Failure in March 2023

In March 2023 a bank with about \$200 billion in assets failed in two days. Its portfolio included one of the five kinds of promises on your list.

## Which one, and why?

The security did not default; every payment that came due was made. Yet the bank still failed.

As you watch, identify which promise the bank held and what changed before it failed. Week 2 gives us a model for checking the explanation.

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**Clip:** [PBS NewsHour, 28 March 2023, 6 minutes](#)

# Summary

Financial decisions start with payment schedules, but the largest payment is not automatically the best choice.

## Key takeaways

- **Markets connect investors and companies.** Cornell invests its endowment; going public lets a company raise capital and its shareholders trade.
- **A product name is a label.** What matters first is the payment schedule: what gets paid, when, and under what conditions.
- **Models earn trust one test at a time.** Theory, backtest, forward test, and real money use different evidence and can reveal different failures.

On Thursday, we build our first comparison rule: how to compare money paid at different times.



## Further Reading

These readings are optional and will not appear on a problem set. Use them if you want more background on the Federal Reserve, monetary policy, or the Treasury.

- *The Structure and Functions of the Federal Reserve System*, Federal Reserve Board: who does what inside the Federal Reserve.
- *Economy at a Glance: Policy Rate*, Federal Reserve Board: how the Fed reports and carries out interest-rate policy.
- *Fedpoints: Federal Reserve System*, Federal Reserve Bank of New York: a short overview of how the system is organized.
- *Role of the Treasury*, U.S. Department of the Treasury: what the Treasury does and how it differs from the Federal Reserve.